

Sentiment analysis for the Uzbek Language Using Classical Machine Learning and XLM-RoBERTa

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I. ABSTRACT

Sentiment analysis for low-resource languages remains a challenging task due to limited annotated corpora, linguistic complexity, and insufficient language-specific NLP resources. Uzbek, despite being spoken by millions of people, remains underrepresented in modern Natural Language Processing research. This study presents a comparative evaluation of classical machine learning methods and transformer-based architectures for Uzbek sentiment analysis.

Four classical machine learning models including Logistic Regression, Naive Bayes, Support Vector Machine (SVM), and Random Forest were evaluated using TF-IDF textual representations. In addition, the multilingual transformer model XLM-RoBERTa was fine-tuned for three-class sentiment classification.

Experiments were conducted using two independent Uzbek sentiment datasets with substantially different class distributions. Classical models achieved strong in-domain validation performance, while XLM-RoBERTa achieved the strongest cross-domain performance on the external test dataset with 45.63% accuracy and weighted F1-score of 0.539.

The results demonstrate that transformer architectures provide stronger semantic generalization under domain mismatch conditions, while classical approaches remain computationally efficient and highly competitive. The study additionally highlights the importance of dataset consistency and domain-aware evaluation for low-resource Uzbek NLP research.

Keywords: Uzbek NLP, Sentiment analysis, XLM-RoBERTa, Machine Learning, Low-Resource Languages, TF-IDF, Transformers

II. INTRODUCTION

Sentiment analysis is one of the most important tasks in Natural Language Processing (NLP). It focuses on identifying emotional polarity and subjective opinions expressed in textual content. Sentiment classification systems are widely used in customer feedback analysis, social media monitoring, recommendation systems, political opinion mining, and business intelligence.

Recent advances in deep learning and transformer architectures have significantly improved sentiment analysis performance for high-resource languages such as English, Chinese, and Spanish. However, low-resource languages continue to

face major challenges due to limited annotated datasets and insufficient language-specific NLP tools.

Uzbek is a Turkic language spoken by more than 35 million people primarily in Uzbekistan and neighboring Central Asian regions. Despite its large speaker population, Uzbek remains critically underrepresented in NLP research. Publicly available Uzbek corpora are limited, and few large-scale sentiment analysis studies have been conducted.

Several linguistic characteristics additionally complicate Uzbek NLP tasks. Uzbek is an agglutinative language where multiple suffixes may attach to a single word stem, creating extremely large vocabulary spaces and severe lexical sparsity. Social media text further introduces spelling inconsistency, slang expressions, transliterated Russian vocabulary, and mixed Latin-Cyrillic script usage.

Traditional bag-of-words methods often struggle under such conditions because they rely heavily on exact lexical overlap. Transformer-based architectures theoretically provide stronger contextual understanding through self-attention mechanisms and multilingual pre-training.

The primary goal of this study is to compare classical machine learning approaches and multilingual transformer models for Uzbek sentiment analysis under low-resource conditions.

The main contributions of this work are summarized as follows:

- Comparative evaluation of four classical machine learning models for Uzbek sentiment analysis.
- Fine-tuning and evaluation of the multilingual XLM-RoBERTa transformer model.
- Investigation of domain shift effects between independent Uzbek sentiment datasets.
- Establishment of baseline results for Uzbek three-class sentiment classification.

III. RELATED WORK

Early sentiment analysis systems primarily relied on lexicon-based techniques and manually engineered linguistic rules. Later, machine learning methods such as Naive Bayes, Logistic Regression, and Support Vector Machines became dominant due to their strong performance on sparse textual representations.

TF-IDF vectorization proved particularly effective for textual classification tasks because it captures important lexical

frequency information while reducing the influence of overly common words.

The introduction of transformer-based architectures revolutionized NLP. BERT demonstrated that large-scale bidirectional contextual pre-training significantly improves downstream language understanding tasks.

Multilingual transformer models such as multilingual BERT (mBERT) and XLM-RoBERTa extended these architectures to support dozens of languages simultaneously.

XLM-RoBERTa is particularly important for low-resource NLP because it was pre-trained on multilingual Common Crawl corpora covering 100 languages including Uzbek.

Compared with high-resource languages, Uzbek NLP research remains relatively limited. Existing Uzbek studies primarily focus on morphology, machine translation, and named entity recognition. Large-scale sentiment analysis research for Uzbek remains underdeveloped.

Most previous Uzbek sentiment analysis studies rely on small datasets and traditional machine learning approaches. Comprehensive comparisons between classical machine learning and multilingual transformers under domain shift conditions remain limited.

This study addresses this gap through systematic evaluation of both classical and transformer-based approaches using independent Uzbek sentiment datasets.

IV. METHODOLOGY

A. Datasets

Two independent Uzbek sentiment datasets were used in this study.

Dataset1 (df1) contains 43,688 textual samples labeled into three sentiment categories:

- Negative
- Neutral
- Positive

The dataset distribution is heavily imbalanced:

- Neutral: 32,578
- Positive: 6,494
- Negative: 4,616

Dataset2 (df2) was used as an external test dataset and contains 9,997 samples:

- Positive: 7,669
- Negative: 1,858
- Neutral: 470

The significant distribution mismatch between datasets creates a strong domain shift problem.

B. Preprocessing

Text preprocessing was carefully designed to preserve important linguistic characteristics of Uzbek.

The preprocessing pipeline included:

- Lowercasing all text
- Removal of excessive punctuation
- Emoji cleaning
- Whitespace normalization

- Removal of malformed symbols

aggressive stemming and stopword removal were intentionally avoided because Uzbek is highly agglutinative. Removing suffix structures may eliminate important semantic and sentiment information.

Social media expressions, slang, and informal vocabulary were preserved to maintain realistic linguistic characteristics.

C. TF-IDF Feature Extraction

For classical machine learning models, TF-IDF vectorization was used.

The vectorizer configuration included:

- Maximum features: 10,000
- N-gram range: (1,2)
- Sublinear TF scaling
- Minimum document frequency threshold

Both unigrams and bigrams were included to capture contextual sentiment expressions.

D. Classical Machine Learning Models

Four classical machine learning models were evaluated:

1) *Logistic Regression*: Logistic Regression was trained using multinomial classification with L2 regularization.

2) *Naive Bayes*: Multinomial Naive Bayes served as a lightweight probabilistic baseline model.

3) *Support Vector Machine*: Linear Support Vector Machine (SVM) was selected because of its strong performance on sparse high-dimensional textual features.

4) *Random Forest*: Random Forest classification used an ensemble of decision trees to improve robustness.

E. Cross-Validation

Five-fold stratified cross-validation was performed on Dataset1.

Cross-validation ensured reliable in-domain evaluation while preserving class distributions across folds.

F. Transformer Model

The transformer-based component of this study used XLM-RoBERTa-base.

XLM-RoBERTa was selected because it provides strong multilingual contextual embeddings and includes Uzbek during pre-training.

The model uses SentencePiece subword tokenization which helps reduce vocabulary sparsity caused by Uzbek morphology.

Fine-tuning configuration included:

- Maximum sequence length: 128
- Learning rate: $2e-5$
- adamW optimizer
- Batch size: 32
- Three epochs

Weighted loss functions were additionally used to reduce class imbalance effects.

G. Evaluation Metrics

The following evaluation metrics were used:

- accuracy
- Precision
- Recall
- Weighted F1-score

Weighted averaging ws selected due to severe dataset imbalance.

V. EXPERIMENTAL RESULTS

A. Cross-Validation Results on Dataset1

Table I presents cross-validation results for classical machine learning models.

TABLE I
CROSS-VALIDATION RESULTS ON DATASET1

Model	Mean accuracy	Standard Deviation
Logistic Regression	0.9059	± 0.0050
Naive Bayes	0.7998	± 0.0021
SVM	0.9632	± 0.0034
Random Forest	0.9802	± 0.0018

Random Forest achieved the highest validation accuracy, followed closely by SVM.

The strong in-domain results demonstrate that lexical patterns are highly informative for Uzbek sentiment analysis.

B. External Evaluation on Dataset2

Table II presents external evaluation results on Dataset2.

TABLE II
EXTERNAL EVALUATION RESULTS ON DATASET2

Model	accuracy	Precision	Recall	F1
Logistic Regression	0.343	0.654	0.343	0.429
Naive Bayes	0.308	0.741	0.308	0.395
SVM	0.387	0.680	0.387	0.473
Random Forest	0.384	0.692	0.384	0.470
XLM-RoBERTa	0.456	0.733	0.456	0.539

XLM-RoBERTa achieved the strongest external performance.

although classical models performed extremely well during cross-validation, their performance decreased substantially under cross-domain evaluation.

This demonstrates the severe domain shift between Dataset1 and Dataset2.

C. Visualization and analysis

Figure 1 compares evaluation metrics across all models.

The figure shows that XLM-RoBERT achieved the strongest weighted F1-score and overall accuracy.



Fig. 1. Evaluation metrics comparison across all models on Dataset2

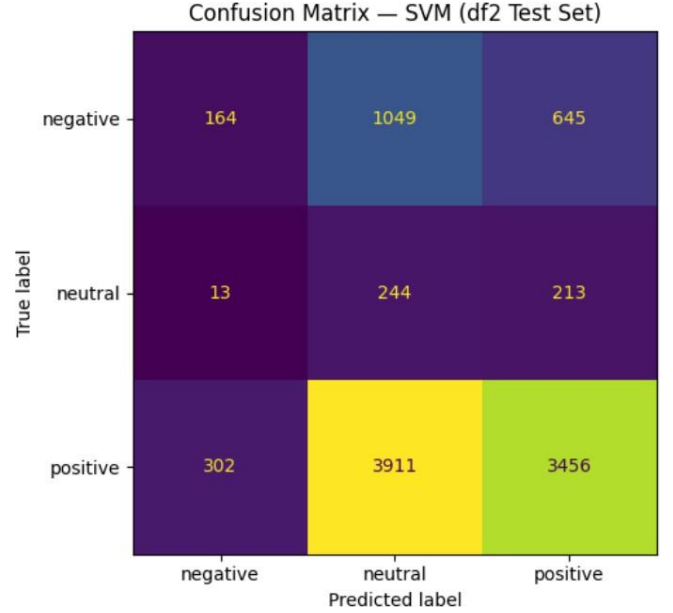


Fig. 2. Confusion matrix for SVM on Dataset2

D. Confusion Matrix analysis

Figure 2 presents the confusion matrix for the best classical model (SVM).

The SVM model strongly over-predicts the neutral class due to Dataset1 imbalance.

Figure 3 presents the confusion matrix for XLM-RoBERTa.

XLM-RoBERTa demonstrates stronger positive sentiment recognition and improved semantic generalization.

E. Training Dynamics

Figure 4 illustrates transformer training loss.

The loss decreased consistently throughout training, indicating stable optimization.

Figure 5 presents validation accuracy and weighted F1-score across epochs.

The transformer model converged rapidly and achieved high validation performance.

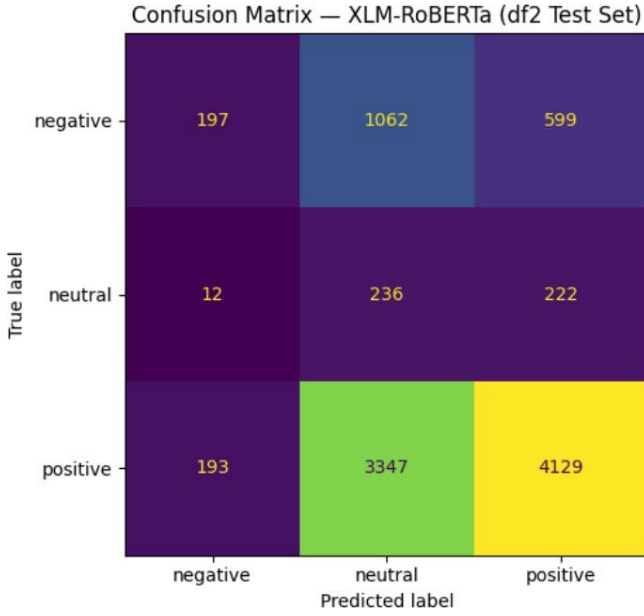


Fig. 3. Confusion matrix for XLM-RoBERTa on Dataset2



Fig. 4. XLM-RoBERTa training loss curve

VI. DISCUSSION

The results demonstrate several important findings regarding Uzbek sentiment analysis.

First, classical machine learning approaches remain highly competitive despite their simplicity. SVM and Random Forest achieved excellent in-domain validation performance while requiring substantially lower computational resources than transformer models.

Second, transformer architectures provide stronger cross-domain semantic generalization. XLM-RoBERTa achieved the highest external accuracy and weighted F1-score because contextual embeddings better handle vocabulary variation and informal language.

Third, domain shift represents the most significant challenge identified in this study. The class distribution mismatch between Dataset1 and Dataset2 substantially reduced model generalization performance.

The dominance of neutral samples in Dataset1 caused models to over-predict neutral sentiment during external evaluation.

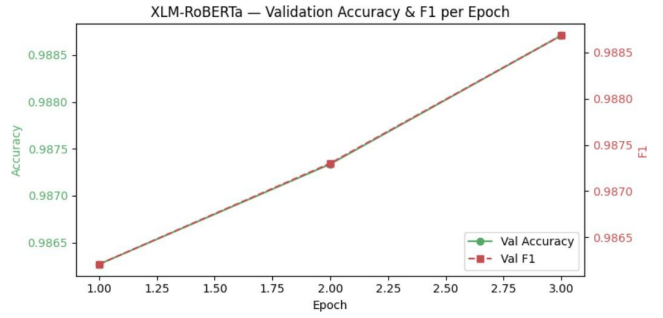


Fig. 5. Validation accuracy and F1-score during XLM-RoBERTa training

another important observation concerns Uzbek morphology. agglutinative suffix structures create large numbers of unique word forms that increase vocabulary sparsity.

Subword tokenization partially mitigated this problem for XLM-RoBERTa.

The experiments additionally revealed that dataset quality may be more important than model complexity alone. Even sophisticated transformer architectures struggle under severe domain mismatch conditions.

A. Computational Complexity

Classical machine learning models were computationally efficient and suitable for CPU-based execution.

Logistic Regression and Naive Bayes required relatively low memory and training time.

SVM training required moderately higher computational cost but remained manageable.

Transformer training, in contrast, required GPU acceleration due to attention computations and millions of trainable parameters.

although computationally expensive, transformer models provided stronger semantic understanding and multilingual transfer capability.

B. Low-Resource Language Challenges

Uzbek NLP introduces several unique challenges.

First, publicly available annotated datasets remain limited.

Second, mixed-script writing and transliterated Russian vocabulary create inconsistent lexical patterns.

Third, social media data frequently contains slang expressions, abbreviations, emojis, and informal grammar.

Fourth, notation consistency between independent datasets remains difficult to guarantee.

These challenges collectively complicate robust sentiment classification.

VII. ERROR ANALYSIS

a deeper investigation of model predictions revealed several recurring error patterns.

One major issue involved neutral sentiment ambiguity. Many Uzbek social media comments contain mixed emotional expressions where positive and negative indicators appear simultaneously. Classical TF-IDF models struggled significantly

under such conditions because sparse lexical features cannot fully capture semantic nuance.

For example, sarcastic or ironic statements frequently caused misclassification. Several comments expressing criticism through humor were incorrectly classified as positive due to the presence of positive lexical indicators.

another major challenge involved code-switching behavior. Uzbek users often combine Uzbek, Russian, and English vocabulary within a single sentence. Classical models heavily depend on exact lexical overlap and therefore failed when transliterated or multilingual tokens appeared.

XLM-RoBERTa handled these cases more effectively because multilingual pre-training exposed the model to multiple languages and cross-lingual semantic patterns.

Spelling inconsistency additionally reduced performance. Informal social media texts frequently contain:

- Character repetition
- Slang abbreviations
- Phonetic spelling
- Missing punctuation
- Latin-Cyrillic mixing

Such variability significantly increases vocabulary sparsity for classical machine learning approaches.

The confusion matrices additionally demonstrate that most errors occurred between neutral and positive classes. This observation suggests that emotional intensity boundaries remain difficult even for transformer models.

VIII. PRACTICAL APPLICATIONS

Uzbek sentiment analysis systems have numerous practical applications across both public and private sectors.

One important application involves social media monitoring. Government institutions and businesses may analyze public sentiment toward policies, services, and products using automated sentiment analysis systems.

E-commerce platforms may additionally use sentiment classification for customer review analysis and recommendation systems.

another promising application concerns public opinion mining during elections and political campaigns. Sentiment analysis may help identify major public concerns and measure emotional responses to political events.

Educational institutions may also benefit from automated analysis of student feedback and online discussion platforms.

In the healthcare domain, sentiment analysis may assist mental health monitoring by identifying emotionally negative patterns in online communication.

The deployment feasibility of such systems strongly depends on computational constraints.

Classical machine learning models remain attractive for lightweight applications because they require relatively low inference time and memory usage.

Transformer models provide stronger semantic understanding but may require cloud infrastructure or GPU acceleration for real-time deployment.

IX. COMPUTATIONAL PERFORMANCE ANALYSIS

The experiments additionally revealed substantial differences in computational efficiency between classical and transformer-based approaches.

Training classical machine learning models required only several minutes on standard CPU hardware.

Naive Bayes and Logistic Regression were especially efficient because of their relatively simple optimization procedures.

SVM training required somewhat higher computational complexity due to margin optimization in high-dimensional feature space.

Random Forest additionally required increased memory consumption because multiple decision trees must be stored simultaneously.

In contrast, XLM-RoBERTa fine-tuning required substantially longer training time.

The transformer model required approximately two hours of training on CPU hardware for three epochs. This illustrates one of the major barriers facing low-resource NLP researchers without access to dedicated GPUs.

However, despite the computational cost, transformer architectures provided superior contextual understanding and stronger external generalization.

The tradeoff between computational efficiency and predictive performance therefore represents an important consideration for practical Uzbek NLP deployment.

X. LIMITATIONS

Several limitations should be acknowledged.

First, the datasets used in this study originate from different domains and contain significantly different sentiment distributions.

Second, no Uzbek-specific morphological analyzers or linguistic normalization tools were incorporated.

Third, transformer training was constrained by available computational resources.

Finally, annotation inconsistencies across datasets may have contributed to reduced cross-domain performance.

XI. CONCLUSION AND FUTURE WORK

This paper presented a comparative study of classical machine learning and transformer-based approaches for Uzbek sentiment analysis.

Experimental results demonstrated that XLM-RoBERTa achieved the strongest external evaluation performance with 45.63% accuracy and weighted F1-score of 0.539.

Classical machine learning approaches nevertheless remained highly competitive while requiring substantially lower computational resources.

The experiments additionally highlighted the severe impact of domain shift and class imbalance on Uzbek sentiment analysis.

Future work should focus on:

- Larger Uzbek sentiment corpora
- Uzbek-specific transformer pre-training

- Domain adaption techniques
- Morphology-aware tokenization
- Better annotation consistency
- Cross-lingual transfer from related Turkic languages

Despite current limitations, this study establishes strong baseline results for Uzbek sentiment analysis and contributes toward future low-resource NLP research.

XII. IMPLEMENTATION DETAILS

This section presents selected implementation snippets used during experimentation.

The preprocessing and TF-IDF vectorization pipeline used for classical machine learning models is shown below.

```
1 vectorizer =
2     TfidfVectorizer( max_featu
3         res=10000,
4         ngram_range=(1,2),
5         sublinear_tf=True
6     )
7
8 X_train = vectorizer.fit_transform(train_texts)
```

The Support Vector Machine classifier configuration is presented below.

```
1 svm_model = LinearSVC()
2 svm_model.fit(X_train, y_train)
3
4 predictions = svm_model.predict(X_test)
```

The transformer-based XLM-RoBERTa implementation used HuggingFace Transformers.

```
1 model = autoModelForSequenceClassification.
2 from_pretrained(
3     "xlm-roberta-base",
4     num_labels=3
5 )
```

The tokenization procedure used during transformer training is shown below.

```
1 train_encodings =
2     tokenizer( train_texts,
3         truncation=True,
4         padding=True,
5         max_length=128
6 )
```

Training was performed using HuggingFace Trainer API.

```
1 trainer =
2     Trainer( model=model,
3         args=training_args,
4         train_dataset=train_dataset, eval_dataset=val_dataset
5     )
6
```

These implementation details demonstrate the practical workflow used throughout the experiments and improve reproducibility for future Uzbek NLP research.

XIII. DATASET DISTRIBUTION DISCUSSION

The distribution imbalance observed across both datasets significantly influenced model behavior throughout the experiments.

Dataset1 contained a strong dominance of neutral samples, while Dataset2 was primarily composed of positive sentiment examples. Such imbalance created a distribution mismatch problem that negatively affected cross-domain generalization.

The imbalance particularly impacted minority negative sentiment recognition. During external evaluation, multiple models demonstrated lower recall for negative samples because the training process was strongly influenced by the dominant neutral class.

This issue is especially problematic for low-resource languages because collecting balanced labeled corpora is often difficult and expensive.

The experiments additionally showed that validation accuracy alone may provide misleading conclusions when datasets originate from substantially different domains.

Several classical models achieved extremely high in-domain validation performance exceeding 96%, yet external accuracy remained below 40%.

This observation highlights the importance of domain-aware evaluation for realistic NLP system assessment.

Transformer architectures partially mitigated these issues through contextual embeddings and multilingual pre-training. However, even XLM-RoBERTa experienced substantial performance degradation under severe domain mismatch.

These findings demonstrate that future Uzbek NLP research should prioritize dataset quality, annotation consistency, and domain diversity alongside model architecture improvements.

XIV. REPRODUCIBILITY AND EXPERIMENTAL ENVIRONMENT

The experiments were conducted using Python 3.11 with the following major libraries:

- Scikit-learn
- Pandas
- NumPy
- HuggingFace Transformers
- PyTorch
- Matplotlib

Classical machine learning experiments were performed on CPU hardware.

Transformer fine-tuning additionally required significant computational resources and longer execution times.

The XLM-RoBERTa training process required approximately two hours on CPU-only execution for three epochs.

To ensure reproducibility, fixed random seeds were used across training and evaluation stages.

Five-fold stratified cross-validation was additionally applied to maintain balanced class distributions during validation.

The experiments demonstrate that meaningful Uzbek NLP research remains possible even under limited computational resources, although transformer training remains computationally expensive.

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